

Supplementary Materials for Neighborhood-Preserving Voronoi Treemaps

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1 ALGORITHM DETAILS

1.1 Binning & Filtering

Before the Voronoi Optimization, binning and filtering are used to reduce the complexity of potentially existing similarity constraints between nodes. The similarity is a float value in the range of $[0.0 - 1.0]$. We utilize binning with 10 equidistant bins of $binWidth = 0.1$. Bin 0.9 holds values $[1 - 0.9]$, bin 0.8 holds values $[0.9 - 0.8]$, etc. We iterate through the sim. constraints sorted from highest to lowest similarity. Each constraint is assigned to its respective $bin^c = \frac{\lceil c * 10 \rceil}{10}$. Once the constraints are binned, we iterate through all bins from the maximum similarity to the minimum similarity. Sim. constraints are aggregated into and saved as the *filtered sim constraints*, until we have reached the *max similarity count* per node, or we have reached an empty bin. The *max similarity count* is an empirically determined value, based on how many neighbors a well-shaped Voronoi cell can have, which in our case is dependent on the weight attribute of *node n*. If a node is larger, it can potentially have more neighbors, but for the datasets we used in this paper, *max similarity count* was in the range of 5 – 10. Although a more complex method could be used to determine the number of bins, such as the Freedman-Diaconis rule or the Sturge's rule, we found that in our experiments, it did not impact the result significantly. However, other ways of determining the bins could be used. We also experimented with bins based on data quantiles, but found that in edge cases, these could lead to unwanted results. Our goal was to create a fully automatic similarity constraint filtering approach. Therefore, the method has to be able to work with hierarchies of very few similarities, similarities where all nodes are similar to each other, and hierarchies where only very few nodes are very similar. Given the range of datasets used in the paper, our simple automatic approach yielded reasonably good results for various types of hierarchies.

Algorithm 1: Similarity Constraints Binning and Filtering

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1  Input: bins=[ $b_1 = []$ ,  $b_{0.9} = []$ , ...,  $b_{0.1} = []$ ], hierarchy node n
2  Output: hierarchy node n
3  function SimilarityBinning( $n$ )
4       $constraints \leftarrow$  Get constraints of  $n$ 
5      forall  $c$  of  $constraints$  do
6           $i = \text{ceil}(c/0.1) * 0.1$ 
7           $bin_i.push(c)$ 
8      end
9       $filteredConstraints \leftarrow []$ 
10     forall  $bin$  of  $bins$  do
11         while  $bin.length \neq 0$  and  $filteredConstraints.length \leq maxSimilarityCount_c$  do
12              $filteredConstraints.push(bin)$ 
13         end
14     end
15      $n.constraints \leftarrow filteredConstraints$ 

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Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: [xx.xxxx/TVCG.201x.xxxxxx](https://doi.org/10.1109/TVCG.201x.xxxxxx)

2 RESULTS OF COMPARATIVE IMPLEMENTATIONS

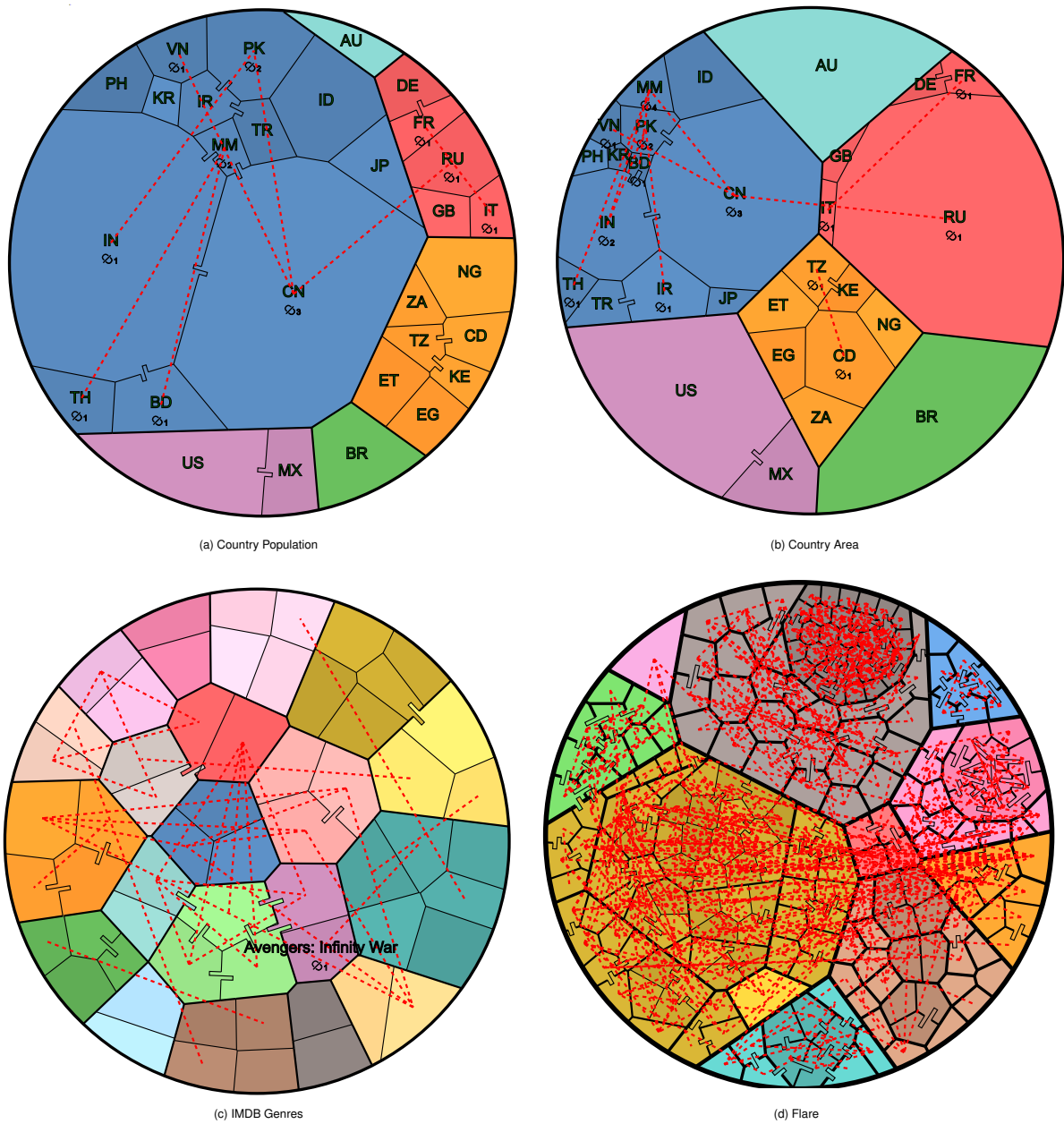


Fig. 1: An overview of results using the CVT approach with random assignment.

3 WINTER OLYMPICS (SHARED SPORT)

This dataset contains different Olympic sports and a selection of three countries that won medals in these sports. As almost all countries have won gold, silver, and bronze medals in many sports, it shows the limitations of our method quite well, as many overlapping cycles exist in the similarity graphs. It also shows the inherent limitations of our work quite well, as the similarity graph contains many cycles, which fight for the same positions, as they have common parents. Therefore, only a minimal number of neighborhoods can be preserved.

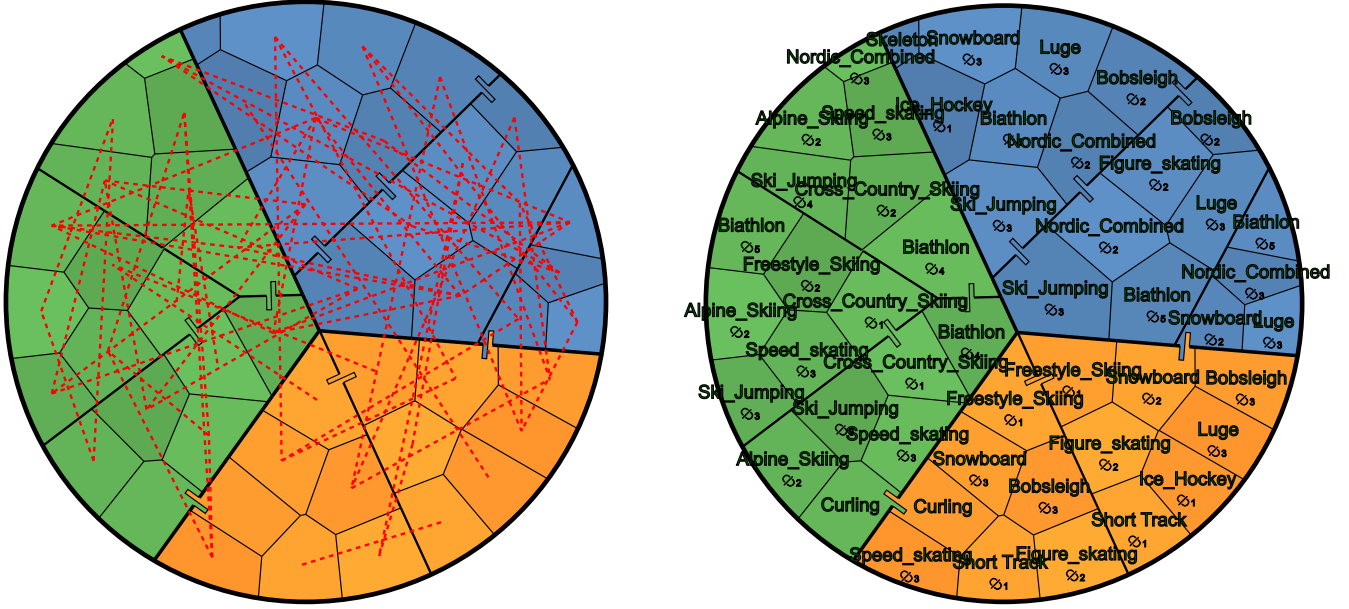


Fig. 3: Olympia Dataset

4 LANGUAGE EVOLUTION (LANGUAGE SIMILARITY)

The dataset comprises similarities across more than 27,000 pairs of languages, encompassing 331 languages [1]. These similarities are derived from the cognates present in the languages' lexicons. In light of the linguistic distribution across Europe [2], we selected six principal language branches and, additionally, languages within each branch that boast a significant number of speakers. Altogether, 35 languages were chosen for this analysis. Utilizing multidimensional scaling (MDS), we compute the embedding for each language based on their mutual similarities. As can be observed in 4, languages within the same parent are highly similar to each other. However, languages such as *French* and *English* are also very similar, even though their word stems differ.

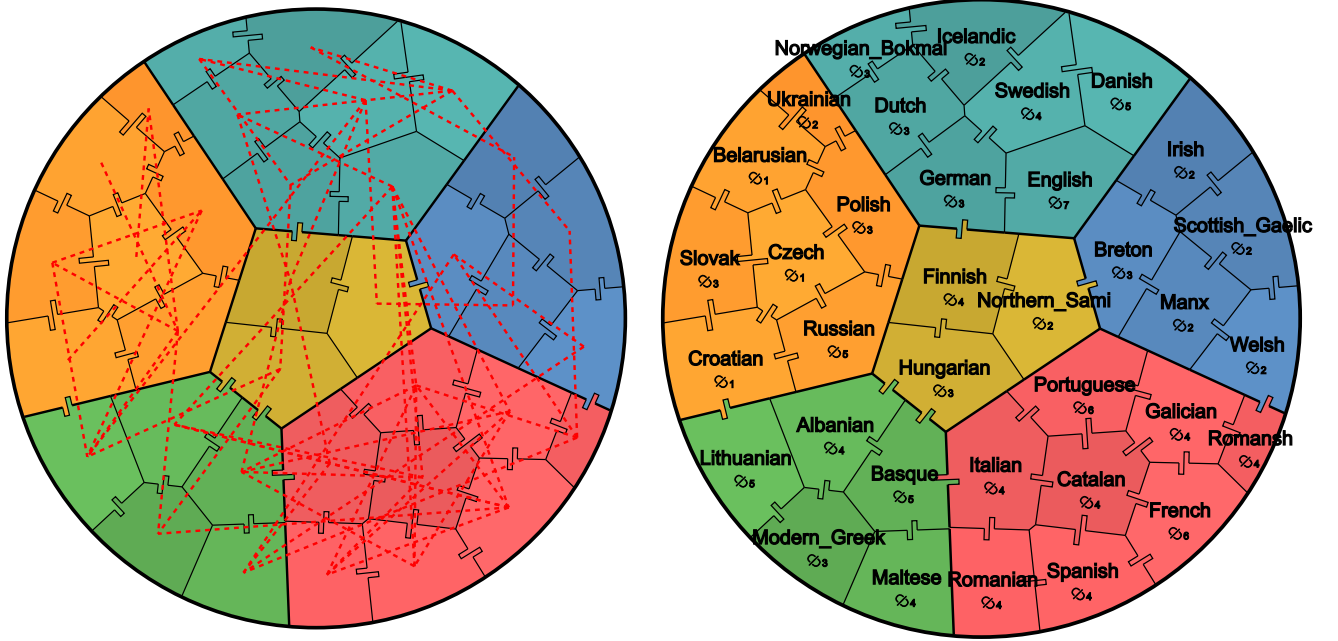


Fig. 4: Language Evolution Dataset

5 SYNTHETIC DATA

Synthetic datasets were used to show how our method handles edge cases, such as full n - n similarities, nodes with many outgoing similarity constraints, and multiple triple similarity constraints. For each example, we show the initialization, as well as the end of the optimization, and the final treemap. Black lines indicate similarity constraints that could not be realized as cell neighborhoods. The green lines indicate similarity constraints that could be realized as cell neighbors. Left: A complex but loosely connected two-level hierarchy. All sibling nodes are connected to each other via similarity constraints, but some nodes share similarity constraints with cousin nodes of another parent. These are correctly positioned after the optimization so that their cells become neighbors. Right: From top to bottom, different numbers of similarity constraints are preserved in simple cases. On the top left, we show how a connectivity where one single node, which has all the similarity constraints, can be resolved. For complex m - n connectivity as shown in the middle, highly connected similarities between siblings can be mostly resolved. In the top right, e.g., where all nodes are connected in a fully-connected way, it is only possible to realize limited similarity constraints as neighborhoods. In general, the number of similarity constraints that can be preserved mostly depends on the number of fully connected, disconnected components of groups of similarity constraints that share parents. Preserving neighborhood relations in datasets that have multiple of these cycles is extremely difficult if they contain cousin relations. This is simply because there is only a single position within each parent cell where such a cycle can be optimally positioned. As we can only position one cycle on a triple intersection point, other cycles cannot realize all similarity constraints as neighborhoods. Therefore, we stop the attraction of cells if they get too close to their parents' boundary or if their target node is already surrounded by other nodes that share similarity constraints. A completely different approach to solving these problematic cycles might be to group them together and display them visually as one cell.

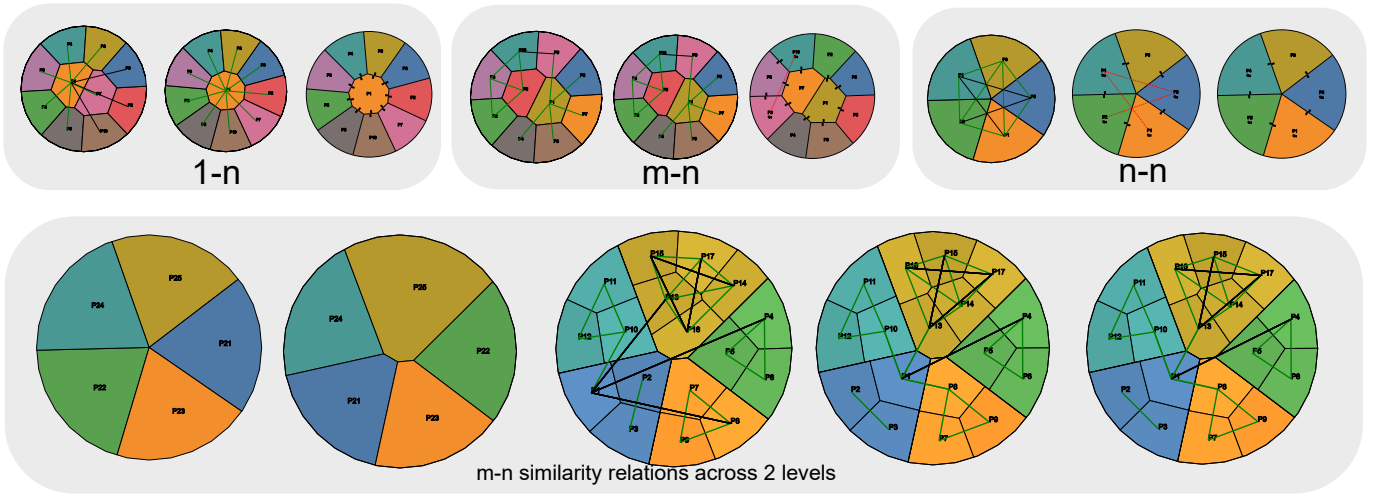


Fig. 5: An overview of different synthetic datasets.

REFERENCES

- [1] G. Bella, K. Batsuren, and F. Giunchiglia. A database and visualization of the similarity of contemporary lexicons. In *International Conference on Text, Speech, and Dialogue*, pp. 95–104. Springer, 2021. 5
- [2] Wikipedia contributors. Languages of europe. https://en.wikipedia.org/wiki/Languages_of_Europe. Accessed: 2024-03-30. 5